

ADA – PLC Guide

As-Built Documentation of the Programmed Ladder Logic

Nebico Oil Melting/Retention System – Version 1.1 (detailed edition)

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This edition folds the V1.0 baseline and the V1.1 changes into one continuous walkthrough, adds a discussion of how the fault system was simplified, and keeps a running list of everything still sitting on a placeholder or a structural guess that needs a real, site-measured value before commissioning.

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1 Purpose and Scope

This guide documents the Nebico Oil Melting/Retention PLC program as it now stands – not as a forward specification – covering the V1.0 baseline and everything V1.1 added, together, in one walkthrough. Three things follow from that:

- Section 3 covers the *entire* program in its current (V1.1) state. Wherever a subsection is new in V1.1, a result box directly underneath says what open question it settles – whether that question came from the V1.0 guide, from re-reading this document later, or from a point *ADA Architecture II* could only speculate about at the time.
- Section 7 looks at *how* the program changed, not just what changed – mainly the fault-detection system, which got simpler in shape and then partly tunable again, and what that trade gives up.
- Section 8 is a running list of every constant in the program standing in for a real physical quantity – a level percentage, a mass conversion, a phase-transition temperature – rather than a structural choice such as a sample count or loop index. These are the values that need a proper, measured number at commissioning, not whatever happens to be sitting in the ladder now.

2 Program Structure Overview

The program's own line-statement index, current as of V1.1. Sections added in V1.1 are bolded.

Step	Section
0	System Install
23	Timer for System
37	MODBUS Logic
240	Automation
243	Physical E-Switch
246	Virtual E-Switches
262	System OK
267	Sensor Average 1
378	Sensor Average 2
487	Water Temperature Target Core Logic
551	Solar Gating
569	Heater Latching
734	Height-to-Mass Conversion
787	Dynamic Upper Thermal Boundary (T5)
853	PO Designation
899	PO Melting Timer
1077	PO Scores
1195	Priority Scheduler
1250	Mode Scheduler
1290	Mode Display in HMI
1314	PO Designated?
1318	PO Score Comparison
1414	Need Array Core
1780	Oil Distribution
2096	Sensor Change Rate Calculations
2287	Fault Analysis (High)
2398	Fault Analysis (Calculations for Low)
2529	Fault Analysis (Low)
2600	Oil Distribution Chart
2674	Jacket Pump
2676	AODD Pump
2749	Manual Oil Pump
2757	HMI Display for Inputs
2821	HMI Charting
2858	Mode Adjuster for Manual Control
2948	END

3 Functional Walkthrough

3.1 System Install (Step 0)

On the first-scan pulse (M8002), the program writes fixed values into a small set of special registers (D8397, D8400, D8401) and sets M8192 – boilerplate configuration of the PLC’s built-in communication port, done once at power-up. V1.1 adds MOV K0 D550 here, initialising the new Modbus write-back register used in §3.20.

3.2 Timer for System (Step 23)

A chained timer sequence off the always-on system bit M8000: T0 (preset K1) gates T1 (preset K5), which gates T2 (preset K60, self-resetting). T1’s contact is the master “logic enabled” permissive that nearly every later section of the program leans on.

Settled by V1.1. An explicit RST T1 now appears at the end of the Sensor Average 1 block, which confirms T1 is a genuinely self-resetting periodic tick – fires, does its work, resets, fires again every K5 – rather than a permanently-true permissive as the V1.0 guide described it.

3.3 MODBUS Logic (Step 37)

A three-station round-robin poll sequencer built on ADPRW. Each of the three stations is polled in two exchanges, gated by timers T10–T61 and stepped by flags M0–M5. The bits of D0, D2, and D4 are unpacked into M51–M59.

This is the real, working version of the satellite mirror word-packing that *PLC Implementation Rev. 2* could only put forward as an unconfirmed guess.

3.4 Automation, Physical E-Switch, Virtual E-Switches, System OK (Steps 240–262)

X001/X000 mirror into M61/M60 (auto/manual selector, physical E-Stop), each driving an indicator lamp. Ten fault bits (M83–M92, populated by Fault Analysis) OR into M62, the virtual E-Stop equivalent. M63 – the permissive nearly every output in the program relies on – is derived from M60, M61, M62, and T1 together, matching Architecture II §12.1’s Physical/Virtual E-Stop → System-OK chain.

3.5 Sensor Average 1 & 2 (Steps 267–486)

Two identical five-sample rolling averages. Block 1: raw inputs D17–D21 → accumulators D100–D108 → filtered D120–D128 (counter D60). Block 2: D22–D26 → D110–D118 → D130–D138 (counter D70). Ten channels filtered in total, matching Architecture II §4.

3.6 Water Temperature Target Core Logic (Step 487)

T_{target} (D150) against level D120: flat at 70.0°C below $d = 200$, flat at 40.0°C above $d = 360$, and a quadratic taper between the two. This confirms Rev. 2 Networks 9–11’s three-region law.

Calibration item. See Section 8 – K200 and K360 are meant to stand for 50% and 90% of the tank’s usable level range, not arbitrary numbers. Real, site-measured values are still owed at commissioning.

3.7 Solar Gating (Step 551)

D150 (T_{target}) versus D136 (filtered solar temperature) routes to Y020 (high side) or Y021 (low side), matching Rev. 2 Network 12.

V1.1 change. Both outputs now also require M63 (SYSTEM_OK) – previously they depended on M200/M205 alone.

3.8 Heater Latching (Step 569)

A short timer pair (T5/T6) samples D120 into a shadow register and compares it via a 32-bit ratio to gate a stability flag; combined with banded compares of D132, this produces a 0–3 heater-stage count (D152) with hysteresis via separate rising and falling branches. D152 drives Y010/Y011/Y012, matching Rev. 2 Network 18’s threshold-count pattern.

3.9 Height-to-Mass Conversion (Step 734)

Four level values scaled by DEMUL ... K1 into mass proxies D160–D166.

Calibration item. The gain sits at unity (K1) across all four channels – a placeholder, not a calibrated height-to-mass conversion. See Section 8.

3.10 Dynamic Upper Thermal Boundary / T5 (Step 787)

T_f via the mass-weighted mixing formula, using constant E239.53.

$239.53 = 479060/2000$ – the exact thermal-capacitance constant from *ADA Architecture II* §5, so the formula went in exactly as designed.

3.11 PO Designation (Step 853)

M12/M13/M14 select a tank’s mass and temperature into working registers D168/D180, matching Rev. 2 Network 19.

3.12 PO Melting Timer (Step 899)

A two-phase demand function (melting/liquid) into D192, matching Rev. 2 §6 (Networks 22–23). M20 sets when $D192 > 0$.

V1.1 addition. A watchdog (T130, gated on T1 AND D192) now clears D192, D16, and M15 on timeout – **T130 is the PO-melt- demand reset timer**. A parallel path driven by **M17 (the main HMI’s reset button)** also clears D16.

Calibration item. E135 and E30 (the melting-point / target comparison constants in this block) are meant to represent real palm-oil phase temperatures and should be checked against the actual product specification. See Section 8.

3.13 PO Scores (Step 1077)

Weighted tank scores (0.6 level / 0.4 temperature) into D212/D214/D216, matching Rev. 2 §9’s weighting, done directly in float.

3.14 Priority Scheduler (Step 1195)

Three condition paths (M250/M251/M252) from banded compares on D128/D16/D120 form the P1/P2/P3-equivalent priority flags.

V1.1 change. M251's condition picked up an M16 term (>D16 K0 AND M20 AND M15 AND M16). **M16 is the pause button at the main HMI** – priority selection now holds off while the process is paused, with help from retentive timer T250.

3.15 Mode Scheduler (Step 1250)

M250/M251/M252 combinations drive Y002/Y003/Y004, latched by M280/M285/M290 – the as-built version of the Need Array trigger-count-to-speed pattern.

3.16 Mode Display in HMI (Step 1290)

Writes a mode-name string into HMI register D480: M290 (LSM) → \$MOV "LSM" D480; M285 (HSM) → \$MOV "HSM" D480; M280 (FSM) → \$MOV "FSM" D480.

New in V1.1 – settles an open point. This section did not exist in V1.0. The source ladder briefly mislabelled the FSM rung as M290 a second time – a plain labelling slip while writing it out rather than a transcription error here – corrected to M280, which matches M280's role as the FSM latch everywhere else in the program.

3.17 PO Designated? (Step 1314)

M270/M271/M272 combine into one aggregate “is a PO tank designated” flag, the as-built equivalent of Rev. 2's ELIGIBLE_COUNT idea.

3.18 PO Score Comparison (Step 1318)

Scores compared pairwise; the running highest and second-highest copy into D218/D220 via chained DEMOVD/D> compares – a max-extraction implementation of Rev. 2 §9.2's incumbent/challenger switching rule.

3.19 Need Array Core (Step 1414)

The largest block in the program: M285/M250/M300 combine with the score comparisons to produce twelve flags (M101–M114), driving the four branch-valve outputs Y011–Y014 alongside M63.

3.20 Oil Distribution (Step 1780)

Three destination request bits push into a real hardware FIFO (SFWR/SFRD), which does away with the manual ring-buffer Rev. 2 had to propose. The dequeued destination drives a dispense-time estimate through a chain that includes DEXP – a genuine exponential instruction, which corrects Rev. 2's assumption that no such instruction existed. The result loads directly as T100's preset.

V1.1 addition – write-back exchange. On each destination request, the program now also sends an ADPRW ... K6 H1 K1 D550 write to the requesting satellite (latches M330/M331/M332, timers T12/T13/T14, K30 preset each, completing on M8029). This is likely aimed at tightening the correlation between HMI and PLC error codes – its exact payload meaning isn’t obvious from the ladder alone, and is worth tracing through properly at some point.

V1.1 addition – oil pump interlock and cancel handling. Y005 now also requires M52, M55, M58 clear, plus a new bit **M50 clear – M50 is the master oil switch at the master HMI**. Separately, M53/M56/M59 (satellite Cancel bits) now reset the computed dispense preset D330 to 0, which was not there in V1.0.

3.21 Sensor Change Rate Calculations (Step 2096)

Roughly every 5 s (T110), the current filtered values snapshot into shadow registers D400–D418, matching Rev. 2’s previous-value shadow register idea.

3.22 Fault Analysis – High / Low (Steps 2287, 2398, 2529)

A ratio of the previous snapshot to the current value is compared against thresholds to set fault bits M83–M92, feeding into the Virtual E-Switch chain.

V1.1 change – see Section 7 for the full discussion. Channels 1–4 now compare against adjustable registers (D500/D502/D504/D506 high-side, D508/D510/D512/D514 low-side) instead of the fixed K2 constant used in V1.0. Channels 5–10 are unchanged.

3.23 Oil Distribution Chart (Step 2600)

Per-destination oil totals accumulate on each completed cycle, with a 1440-minute (24 h) rollover driven by the M8014 minute pulse – a real, working version of Rev. 2’s speculative “T24H” placeholder.

3.24 Jacket Pump, AODD Pump, Manual Oil Pump (Steps 2674–2756)

The jacket pump runs unconditionally on M63 (Rev. 2 Network 45, confirmed exactly). The AODD pump (Y007) is tied to PO tank designation plus a level/temperature threshold against K30, closing an open item Rev. 2 had to carry forward twice. Manual pushbutton paths (M600/M650) give an operator override that sits independently of the automatic logic.

V1.1 change. The Manual Oil Pump section’s Y005 rung picked up the same M52/M55/M58/M50 interlock added in Oil Distribution above.

3.25 HMI Display for Inputs (Step 2757)

A direct 32-point mirror of the physical digital inputs: X000–X037 → M115–M146.

New in V1.1 – settles an open point. This did not exist in V1.0. It confirms, as a real block, the “32-point input detection mirror” idea *ADA Architecture II*’s HMI document could only speculate about (there numbered M40–M71). **This block fully occupies M115–M146** – any free-coil allocation planned against that range needs to move to M147–M152 or elsewhere.

3.26 HMI Charting (Step 2821)

Four values copy into a dedicated trend-chart block once per T1 tick: DEMOVD178 D6 (T_f), DEMOVD120 D7 (T_5 level), DEMOVD150 D8 (T_{target}), DEMOVD138 D9 (channel-10 temperature).

New in V1.1. Did not exist in V1.0; exists purely to feed an HMI trend/history chart from a stable, separate register block rather than reading the live process registers directly.

3.27 Mode Adjuster for Manual Control (Step 2858)

A parallel manual-override path for the three CP speed outputs: M8–M11 all set, together with M60/M61/M62 and a new coil M610, drive Y002/Y003/Y004 directly, sitting alongside the automatic Need Array path.

New in V1.1 – settles an open point. Did not exist in V1.0. **M8–M11 and M610 are the manual mode-adjuster inputs** – an HMI/panel-driven manual override sitting alongside, not replacing, the automatic speed-mode logic documented above.

4 Timer/Counter Presets

Device	Preset	Role
T0	K1	Power-up settling delay
T1	K5	Master logic-enabled tick (self-resetting, settled §3)
T2	K60	Self-resetting housekeeping pulse
T10, T20, T30, T40, T50, T60	K1/K2	MODBUS sequencer, stations 1–3
T11, T21, T31, T41, T51, T61	K1/K2	MODBUS sequencer, secondary exchange
T5	K3000	Heater-latch sample timer
T6	K3030	Heater-latch sample timer
T100	D330 (calculated)	Oil dispense duration – register preset, not a fixed constant
T110	K50	Sensor change-rate sample tick (~ 5 s)
T12	K30	V1.1: write-back exchange timeout, station 1
T13	K30	V1.1: write-back exchange timeout, station 2
T14	K30	V1.1: write-back exchange timeout, station 3
T130	TBD	V1.1: PO-melt-demand reset timer – role settled, preset value not present in this export

5 Reconstructed Register Map

Register range	Inferred role
D6–D9	(V1.1) HMI trend-chart mirror
D16	Priority-scheduler condition register, cleared by T130/M17
D17–D26	Raw sensor inputs (10 channels)
D60, D70	Sample counters, sensor averaging
D100–D118	Rolling-sum accumulators
D120–D138	Filtered sensor values, 10 channels
D140, D148	Heater-latch stability sample/ratio
D150	T_{target}
D152	Heater stage count, 0–3
D160–D166	Mass proxies (unity gain – calibration item)
D168, D180	Selected PO tank working temperature/mass
D170–D178	T_f calculation
D182–D192	PO melting/liquid demand function
D200–D216	Weighted tank scores
D218, D220	Running highest/second score
D300	Oil destination FIFO
D305	Dequeued active destination
D310–D330	Dispense quantity / runtime-estimate chain
D400–D418	Previous-value shadow registers
D420–D438	Rate-of-change ratios
D440–D458	Low-side ratio checks
D460–D462	Cumulative oil consumption
D470	Running minute counter (24 h rollover)
D480	(V1.1) HMI mode-name string display
D500, D502, D504, D506	(V1.1) Adjustable fault high-thresholds, channels 1–4
D508, D510, D512, D514	(V1.1) Adjustable fault low-multipliers, channels 1–4
D550	(V1.1) Modbus write-back value to active satellite
M0–M5	MODBUS sequencer state
M12–M14	PO tank select
M16	(V1.1) Pause button, main HMI
M17	(V1.1) Reset button, main HMI
M51–M59	Unpacked satellite mirror bits
M60–M63	Physical/Virtual E-Switch, System OK
M83–M92	Fault bits, 10 channels
M100–M114	Heater/priority/scheduler intermediate flags
M115–M146	(V1.1) Physical input mirror, X000–X037
M200–M230	Heater hysteresis and stability latches
M250–M300	Priority/mode scheduler latches
M270–M272	PO designation latches
M323, M324	Oil FIFO dequeue handshake
M330, M331, M332	(V1.1) Write-back exchange latches
M470	Daily total reset pulse
M8–M11	(V1.1) Manual mode-adjuster inputs
M50	(V1.1) Master oil switch, master HMI
M610	(V1.1) Mode-adjuster gating flag

6 Cross-Reference to ADA Architecture II / PLC Implementation Rev. 2

Rev. 2 item	As-built status
Satellite mirror word packing (Network 39, unconfirmed guess)	Confirmed. D0/D2/D4 bit-unpacking implements it
32-point input detection mirror (HMI doc, M40–M71)	Confirmed, V1.1. As-built at M115–M146
Oil destination FIFO (manual ring buffer, Networks 39–40)	Superseded. Real SFWR/SFRD hardware FIFO
Exponential block for viscosity correction (“not available”, Network 52)	Incorrect assumption. DEXP exists and is used
$T_{\text{target}}(d)$ three-region law (Networks 9–11)	Confirmed, same shape
T_f mass-weighted boundary, $K = 479060$ (Network 20)	Confirmed exactly
Tank scoring, 0.6/0.4 weighting (§9)	Confirmed, in float
24-hour oil consumption rollover, “T24H” placeholder (Network 44)	Resolved. Real M8014 minute-pulse counter
AODD pump duty, unresolved (carried in Rev. 2 twice)	Resolved. Tied to PO designation + threshold
Trend Threshold Supervision, per-channel rise/fall registers (§12.2)	Simplified, then partly restored. See Section 7
Manual override provisions (not specified in Rev. 2)	Added, V1.1. Mode Adjuster (§3)
Height-to-mass calibration	Not yet calibrated – unity gain placeholder

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7 Changes Implemented in Programming

7.1 Fault System: Simplified, Then Selectively Restored

ADA Architecture II §12.2 called for a per-channel Trend Threshold Supervision with separate, individually-adjustable rise and fall limit registers across all twelve channels (six tank/mass-or-temperature pairs) – twelve distinct limit registers, each meant to be tunable by the operator from day one.

What I actually built took a different path. Fault Analysis (High/Low, §3) collapses the twelve-channel, rise/fall- separated design into **one ratio check per channel, read from both directions**: a single DEDIV of the previous value against the current one, checked against a high-side threshold and, separately, a doubled version of that same ratio checked against a low-side threshold. Ten channels, ten fault bits, two comparisons each – structurally much simpler than the architecture’s twelve-register design, and originally (V1.0) hardcoded entirely to K2/K1 constants with no runtime adjustment at all.

V1.1 then brings back some of that tunability – but not by returning to the architecture’s twelve-register-per-channel design. Only the **first four channels** (the T5-level and three PO-tank-level channels) had their K2/K1 constants swapped for live registers (D500–D514, Section 7.2). The remaining six temperature-derived channels are still fixed.

The trade-off, plainly. The architecture’s original twelve-register design gives full, symmetric, per-channel tunability at the cost of complexity – more registers to keep track of, more surface area for a commissioning mistake, more scan-time and program-step overhead. What I built instead is simpler and more reliable to write and maintain – fewer registers, fewer comparison branches, one pattern repeated ten times – but gives up per-direction independence: a single ratio threshold governs both directions per channel, and only the four level channels can be tuned at all without a program change. That is a sensible trade for a first commissioning pass – simpler logic is easier to check and less likely to hide a bug – but it is a deliberate narrowing of the architecture’s original scope, not a like-for-like build of it.

7.2 Updated Fault Threshold Registers

Channel	High threshold	Low multiplier	Status
1 (D420, from D120 – T5 level)	D500	D508	Adjustable, V1.1
2 (D422, from D122 – TA level)	D502	D510	Adjustable, V1.1
3 (D424, from D124 – TB level)	D504	D512	Adjustable, V1.1
4 (D426, from D126 – TC level)	D506	D514	Adjustable, V1.1
5–10 (D428–D438, temperature channels)	K2	K2	Still fixed

Worth deciding at commissioning: finish the rollout by registerising the remaining six temperature channels too, or treat the current 4-of-10 split as the intended final state. Both are defensible – this is simply a decision still waiting to be made.

7.3 Other Programming Changes Worth Noting

- **Added interlocks, not removed ones.** Every V1.1 change to an existing rung *added* a condition (M63 on solar valves, M50/M52/M55/M58 on the oil pump) – none loosened an existing permissive. The overall direction of this update is more cautious, not less.
- **A second communication exchange, not a redesign.** The write-back to D550 (§3.20) adds a new outbound message per destination request but does not touch the FIFO/dequeue logic it sits alongside – it is additive, not a rework of the oil distribution mechanism.
- **HMI-facing additions outnumber process-logic changes.** Three of the four new sections (Mode Display, HMI Display for Inputs, HMI Charting) exist purely to give the HMI better visibility into the process – they read from existing registers and write to new ones, without altering any control decision. Only the Mode Adjuster section adds new *control* capability, in the shape of a manual speed override.

8 Calibration Items

Everything below is a constant currently sitting in the ladder that stands in for a real, physical, site-specific value. None of these should be treated as final without checking them at commissioning.

Location	Constant(s)	What it needs to become
Water Temperature Target Core (§3)	K200, K360	K200 is meant to stand for 50% of the tank's usable level range; K360, 90%. Both are placeholders until the real level-range measurement is taken at commissioning.
Height-to-Mass Conversion	K1 (DEMUL gain, $\times 4$ channels)	Currently unity gain – needs the real height-to-mass conversion factor per tank, worked out from tank geometry and product density.
PO Melting Timer	E135, E30	Melting-point / target-temperature reference constants – worth checking against the actual palm-oil product specification rather than the values currently in the ladder.
Heater Latching	K55, K80, K95, K85, K70, K45 (banded thresholds on D132)	Heater-stage transition points – confirm these match the intended comfort/staging bands for the actual tank and product.
Heater Latching – stability gate	T5/T6 (K3000/K3030), E1 (DECMPI148 threshold)	Sampling window and stability-ratio threshold – may need adjusting once the real sensor noise characteristics are known.
Fault Analysis, channels 5–10	K2 (high), K1 (low multiplier)	Still hardcoded (Section 7.2) – decide whether these need registering like channels 1–4, or are fine left fixed.
AODD Pump	K30 (level/temperature compare, D132/D134/D130)	Trigger threshold for AODD pump engagement – confirm against real tank levels and temperatures, not the placeholder value.
Oil write-back exchange	T12/T13/T14 (K30 timeout)	Worth confirming this timeout window is realistic for the actual Modbus round-trip time to the satellites, not just a placeholder.
PO Melting demand	D190 (calibration multiplier, already a register)	Already tunable without a program change – but the <i>value</i> loaded into it at commissioning is still an open question.

Not calibration items – structural constants, left as-is. K5 (sample count, sensor averaging), K50 (sensor change-rate tick interval), E239.53 (= 479060/2000, confirmed exactly against Architecture II §5), and the MODBUS sequencer's timer chain (T0/T1/T2, T10–T61) are structural or timing choices rather than stand-ins for a physical measurement, so they are left out of the table above.

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